

Astaroth: A scientific computing framework for accelerating stencil computations.

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Summary

Stencil computations¹ are one of the bedrocks of high-performance scientific computing, forming the core of many partial differential equation (PDE) and numerical linear algebra solvers. In recent years, GPUs have become the primary compute platform for data-parallel applications in high-performance computing, and it has become difficult to run large simulations without them. Astaroth is a GPU framework for stencil computations that has been developed to address this problem of scalable scientific computing. Astaroth provides its own domain-specific language (DSL), in which researchers can express such computations without having to focus on technical implementation details. It can run efficiently in both CUDA- and HIP-based environments, and for testing purposes on CPUs. While stencils are the core of Astaroth, it also accelerates other operations like reductions (e.g. sums), simple ray-tracing, and integrates with libraries performing GPU-accelerated Fourier transforms, all of which are important for simulations on structured grids. Astaroth is optimized for multiphysics use cases and has primarily been used for turbulent astrophysical plasma simulations.

Statement of need

Due to the ubiquity and expense of stencil computations, a common library for implementing them efficiently on GPUs is needed, more so because high-performance codes require portability across different computing platforms. Such a library is both needed for accelerating existing codes and writing new ones. Thus, its API has to enable good integration with existing code and easy development of new code.

Astaroth strives to be this library, by creating abstractions of common stencil structures and operations in its domain-specific language (DSL)², which enables researchers in different domains to write their stencil computations in a compact and clear way. These abstractions allows Astaroth to provide the needed execution platform for different application cases, predominantly for partial differential equation (PDE) solvers, but also for image processing and numerical linear algebra.

In particular, multiphysics simulations include computationally challenging stencil calculations due to the need to hold values of many fields in working memory at the same time. Accelerating them was the original need for which Astaroth was developed. As an example, Astaroth has

¹Stencil computations are computations on structured grids where a given point is updated using a fixed neighborhood pattern. Examples are convolutions in image processing and convolutional neural networks, and different schemes for spatial derivatives like the finite-difference method.

²For a simple example of the Astaroth DSL and a program using it, see the advection solver example from this [link](#).

been employed to accelerate the widely used astrophysics framework Pencil Code, for which existing acceleration methods were not fit.

State of the field

Several approaches have been proposed to improve the performance, portability, and (user) productivity of stencil computations. Single-process approaches include DSLs, e.g., Halide (Ragan-Kelley et al., 2013), PolyMage (Mullapudi et al., 2015), Delite (Sujeeth et al., 2014), and Lift (Steuwer et al., 2017), and general parallel processing abstractions, e.g., Kokkos (Trott et al., 2021) and RAJA (Beckingsale et al., 2019). The primary benefit of a DSL is that assumptions about the structure of computations can be made to improve the performance of the generated code while maintaining a high-level representation for the user. Performance-portability is typically augmented with automated tuning and algorithm selection (e.g., PATUS (Christen et al., 2011), PARTANS (Lutz et al., 2013)). These approaches are also adopted in Astaroth, which introduces a DSL, a code generator, and implements automated thread block size optimizations. A distinctive feature of Astaroth is its algorithmic specialization for cache-constrained use cases in multiphysics, where the working set required to update interdependent physical fields is too large to fit into on-chip caches. This is addressed by reordering the computations into stages of higher and lower cache reuse, thereby enabling better utilization of on-chip caches (Pekkilä et al., 2025). Astaroth not only considers stencils in isolation, but also their combinations with other operations inside the same kernel, such as distributed reductions.

Methods to improve the performance-portability-productivity dimensions in multiprocess applications include topology-aware workload distribution and abstractions for distributed operations. Chapel (Callahan et al., 2004) and Charm++ (Kalé & Krishnan, 1993) improve these dimensions by providing general-purpose distributed programming models, whereas Cactus (Goodale et al., 2003) and PETSc (Mills et al., 2021) implement scientific computation toolkits. Further productivity advances can be gained with domain-specialized frameworks, which provide ready-made solutions for specific use cases and a simplified application programming interface. Astaroth also belongs to this class of frameworks. Examples closest to Astaroth are Parthenon (Grete et al., 2023) and AMReX (Zhang et al., 2019). Both are frameworks for distributed adaptive mesh refinement (AMR), where Parthenon uses Kokkos as the compute backend and AMReX provides compute features with parallel function wrappers and user-written C++ lambdas. In contrast to these projects, Astaroth's distributed abstraction layer focuses on structured grid computations without mesh refinement, and can thus make simplifying assumptions about the underlying data-movement patterns to better address on-chip data movement, kernel fusion, and data-processing pipelines. Similar to Parthenon and others (Pearson, 2021), Astaroth implements its version of topology-aware domain decomposition and rank reordering for improved portability across systems, and performs fused packing to alleviate communication overheads. Furthermore, Astaroth implements a task scheduler for compute and communication tasks (Lappi, 2021).

In the field, Astaroth stands out as a CUDA/HIP stencil-computing framework focused on addressing the performance-productivity trade-off in cache-heavy multiphysics applications with its DSL. This enables holistic optimizations of complex multiphysics simulations and allows for experimentation with optimization techniques across traditionally decoupled tasks (e.g., extensive kernel fusion of operations across the stack), which would not be practical with libraries utilizing opaque submodules for computation and communication.

Software design

Astaroth consists of three main components: 1) acc, a compiler and runtime system for a domain-specific language (DSL) for stencil computations, 2) an API for executing stencil

applications on multi-GPU platforms, and 3) a standalone PDE solver. Below, we present a quick overview of these components. Extensive documentation is available ([The Astaroth developers, 2026](#)).

acc compiler and runtime system

Astaroth has a DSL for stencil-based computations, designed to be used by domain scientists without having to consider technical implementation details. The main operations, such as stencils, are written in a declarative syntax, and the kernels that use them are written in an imperative syntax.³ Their implementation is left to Astaroth's DSL compiler *acc*, which applies a number of specialized optimizations. An especially important one is the unrolled and reordered computation of all required stencils at the start of the kernels, which enables instruction-level parallelism and efficient usage of caches ([Pekkilä, 2026](#)). In addition to stencils, the DSL supports two other operations: 1) multi-GPU reductions – which are commonly needed for stencil-based solvers; and 2) simplified distributed ray-tracing for simulations incorporating radiative transfer ([Heinemann et al., 2006](#)), with the restriction that the rays keep their directions and move only through neighbouring grid points. Astaroth's DSL also includes a standard library, providing, among others: derivative operators used in PDE solvers, implemented for arbitrarily spaced Cartesian, spherical or cylindrical grids; and Poisson solvers, e.g. for self-gravity ([Krasnopolsky et al., 2026](#)).

acc transpiles the DSL source into CUDA or HIP source code, which is further compiled into machine code using a native CUDA or HIP compiler. The program thus produced is executed in the *acc* runtime system, which further optimizes the kernels by autotuning the thread block sizes for kernel execution. *acc* also supports run-time compilation, because run-time configuration parameters may change the evaluation of conditional statements, thereby changing the branches taken at run-time. With run-time compilation, *acc* compiles for a given configuration only those parts of the DSL source that will be executed. Information about which code is executed also allows Astaroth to optimize run-time behaviour more precisely, e.g. memory allocations or communication patterns.

Multi-GPU runtime system and API

In the DSL, using the *ComputeSteps* language construct, users can define a list of compute steps specifying a sequence of kernels and boundary conditions. Kernels defined in *ComputeSteps* may be fused to reduce memory reads. Based on the decomposition of the overall domain into subdomains and the stencils' data access patterns, *acc* infers the dependency relationships between the steps, and constructs a directed acyclic graph (DAG) of dependent tasks. Each step is split into many tasks, one per region of the subdomain: there are 26 smaller regions at each subdomain's boundaries, which are dependent on communicated data from neighbors; and one big region at the core of the subdomain, which is not. Communication tasks are inserted where needed.

Astaroth's task scheduler executes these DAGs, asynchronously launching computation and communication tasks when prerequisite tasks are completed. This improves performance in communication-bound cases, especially for higher process counts ([Lappi, 2021](#)). For fast data transfers and to support all possible hardware, both GPU-to-GPU remote direct memory access (RDMA) and routing of the communication through CPUs are supported.

This runtime system can be accessed through Astaroth's runtime API. The API is C-ABI compatible, supporting function interfaces to external applications written in any programming language. The API is organized into two layers: the Device layer and the Grid layer. The Device layer provides access to single-GPU functionality, such as: moving data between CPU and GPU, launching kernels, and loading/storing snapshots from/to disk. The Grid layer

³In declarative programming, computations are defined by describing what the results look like; in imperative programming, by describing the steps to perform.

provides access to multi-GPU functionality, such as: executing DAGs, distributed initialization, and distributed loading/storing of snapshots. Other special functionality is also provided through the API, such as distributed Fourier transforms.

Solver

Astaroth also includes a standalone finite-difference PDE solver (Pekkilä et al., 2022), which takes full advantage of the DSL and the multi-GPU API, and can be extended to new simulation models. The solver scales to thousands of GPUs with a weak-scaling efficiency greater than 90% (Pekkilä, 2026) and also works as a testbed for performance research.

The solver uses an astrophysical magnetohydrodynamical setup (`acc-runtime/samples/mhd_modular`) by default, but can be configured to run any DSL code. The samples directory also includes other production-ready setups, e.g. `tfm-mpi` for the test-field method (Pekkilä, 2026).

The solver handles distributed initial conditions, domain decomposition, simulation diagnostics, and logging. It is also designed to react to a number of events, such as NaNs in the simulation data, simulation time limits, and a stop signal given through the file system. The directory `analysis/` contains Python-based data analysis tools, which can be used to process the data produced by the standalone solver.

Research impact statement

Astaroth has already been used in a number of papers as the core PDE solver, mainly for astrophysical plasma simulations (Gent et al., 2026; M. S. Väisälä, 2017; Miikka S. Väisälä et al., 2021, 2023), but also in seismology (Ladino et al., 2025). Additionally, it has been used for research on performance optimization methods (Pekkilä et al., 2017, 2025; Pekkilä, 2026), communication techniques (Lappi, 2021; Pekkilä et al., 2022), compiler techniques (Pekkilä, 2019; Puro, 2023) and other topics (Puro & Pohjonen, 2025; Yokelson et al., 2024). We expect that the recent GPU-acceleration of Pencil Code, which was done by embedding Astaroth's DSL and runtime system into it, will increase the number of Astaroth users. The associated speedup factor of 20-60 (Pekkilä et al., 2022) will enable more realistic astrophysical simulations in a wide range of use cases from modelling small-scale dynamos (Warnecke et al., 2025) to processes producing primordial gravitational waves and their propagation (Roper Pol et al., 2020).

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AI usage disclosure

AI tools have not been used for producing documentation or in the authoring of this paper. For software development, AI tools have been used for debugging and generating small snippets of code, always validated by human review.

References

- Beckingsale, D. A., Burmark, J., Hornung, R., Jones, H., Killian, W., Kunen, A. J., Pearce, O., Robinson, P., Ryujin, B. S., & Scogland, T. R. (2019). RAJA: Portable performance for large-scale scientific applications. *2019 IEEE/ACM International Workshop on Performance, Portability and Productivity in HPC (P3hpc)*, 71–81. <https://doi.org/10.1109/p3hpc49587.2019.00012>
- Callahan, D., Chamberlain, B. L., & Zima, H. P. (2004). The Cascade High Productivity Language. *Proceedings of the Ninth International Workshop on High-Level Parallel Programming Models and Supportive Environments, 2004.*, 52–60. <https://doi.org/10.1109/HIPS.2004.1299190>
- Christen, M., Schenk, O., & Burkhart, H. (2011). PATUS: A code generation and autotuning framework for parallel iterative stencil computations on modern microarchitectures. *Proceedings of the 2011 IEEE International Parallel & Distributed Processing Symposium*, 676–687. <https://doi.org/10.1109/IPDPS.2011.70>
- Gent, F. A., Mac Low, M.-M., Korpi-Lagg, M. J., Puro, T., & Rheinhardt, M. (2026). Asymptotic behaviour of galactic small-scale dynamos at modest magnetic Prandtl number. *The Astrophysical Journal Letters*, 1000(2), L40. <https://doi.org/10.3847/2041-8213/ae4c41>
- Goodale, T., Allen, G., Lanfermann, G., Massó, J., Radke, T., Seidel, E., & Shalf, J. (2003). The Cactus Framework and Toolkit: Design and Applications. In G. Goos, J. Hartmanis, J. Van Leeuwen, J. M. L. M. Palma, A. A. Sousa, J. Dongarra, & V. Hernández (Eds.), *High performance computing for computational science — VECPAR 2002* (pp. 197–227). Springer. https://doi.org/10.1007/3-540-36569-9_13
- Grete, P., Dolence, J. C., Miller, J. M., Brown, J., Ryan, B., Gaspar, A., Glines, F., Swaminarayan, S., Lippuner, J., Solomon, C. J., Shipman, G., Junghans, C., Holladay, D., Stone, J. M., & Roberts, L. F. (2023). Parthenon—a Performance Portable Block-Structured Adaptive Mesh Refinement Framework. *The International Journal of High Performance Computing Applications*, 37(5), 465–486. <https://doi.org/10.1177/10943420221143775>
- Heinemann, T., Dobler, W., Nordlund, Å., & Brandenburg, A. (2006). Radiative transfer in decomposed domains. *Astronomy & Astrophysics*, 448(2), 731–737. <https://doi.org/10.1051/0004-6361:20053120>
- Kalé, L. V., & Krishnan, S. (1993). CHARM++: A portable concurrent object oriented system based on C++. *Proceedings of the Eighth Annual Conference on Object-Oriented Programming Systems, Languages, and Applications*, 91–108. <https://doi.org/10.1145/165854.165874>

- Krasnopolsky, R., Puro, T., Li, W.-W., Shang, H., Väisälä, M. S., Mac Low, M.-M., Rheinhardt, M., & Korpi-Lagg, M. (2026). Iterative Poisson Solvers for Self-gravity with the GPU Code Astaroth. *arXiv Preprint arXiv:2605.03563*. <https://doi.org/10.48550/arXiv.2605.03563>
- Ladino, O., Coelho, B. S., Wyzykowski, A. B. V., & Soares, L. (2025). Acoustic Wave Modeling Using the Astaroth Framework. *Empowering the Energy Shift: The Role of HPC in Sustainable Innovation, 2025*, 1–3. <https://doi.org/10.3997/2214-4609.2025643013>
- Lappi, O. (2021). *A Task Scheduler for Astaroth, the Astrophysics Simulation Framework* [Master's thesis, Aalto University]. <https://urn.fi/URN:NBN:fi-fe2021050428868>
- Lutz, T., Fensch, C., & Cole, M. (2013). PARTANS: An autotuning framework for stencil computation on multi-GPU systems. *ACM Transactions on Architecture and Code Optimization*, 9(4), 59:1–59:24. <https://doi.org/10.1145/2400682.2400718>
- Mills, R. T., Adams, M. F., Balay, S., Brown, J., Dener, A., Knepley, M., Kruger, S. E., Morgan, H., Munson, T., Rupp, K., Smith, B. F., Zampini, S., Zhang, H., & Zhang, J. (2021). Toward Performance-Portable PETSc for GPU-Based Exascale Systems. *Parallel Computing*, 108, 102831. <https://doi.org/10.1016/j.parco.2021.102831>
- Mullapudi, R. T., Vasista, V., & Bondhugula, U. (2015). Polymage: Automatic optimization for image processing pipelines. *ACM SIGARCH Computer Architecture News*, 43(1), 429–443. <https://doi.org/10.1145/2694344.2694364>
- Pearson, C. (2021). *Movement and Placement of Non-Contiguous Data in Distributed GPU Computing* [PhD thesis, University of Illinois at Urbana-Champaign]. <https://hdl.handle.net/2142/110511>
- Pekkilä, J. (2019). *Astaroth: A library for stencil computations on graphics processing units* [Master's thesis, Aalto University]. <https://urn.fi/URN:NBN:fi:aalto-201906233993>
- Pekkilä, J. (2026). *Graphics processors in high-performance computing: Practice and experience in the simulation of astrophysical magnetohydrodynamics and turbulence* [PhD thesis, Aalto University School of Science]. <https://urn.fi/URN:ISBN:978-952-64-3160-4>
- Pekkilä, J., Lappi, O., Robertsén, F., & Korpi-Lagg, M. J. (2025). Stencil computations on amd and nvidia graphics processors: Performance and tuning strategies. *Concurrency and Computation: Practice and Experience*, 37(12-14), e70129. <https://doi.org/10.1002/cpe.70129>
- Pekkilä, J., Väisälä, M. S., Käpylä, M. J., Käpylä, P. J., & Anjum, O. (2017). Methods for compressible fluid simulation on GPUs using high-order finite differences. *Computer Physics Communications*, 217, 11–22. <https://doi.org/10.1016/j.cpc.2017.03.011>
- Pekkilä, J., Väisälä, M. S., Käpylä, M. J., Rheinhardt, M., & Lappi, O. (2022). Scalable communication for high-order stencil computations using CUDA-aware MPI. *Parallel Computing*, 111, 102904. <https://doi.org/10.1016/j.parco.2022.102904>
- Puro, T. (2023). *Programmatic and efficient GPU acceleration of CPU stencil codes* [Master's thesis, University of Helsinki]. <http://hdl.handle.net/10138/569388>
- Puro, T., & Pohjonen, A. (2025). GPU acceleration of average gradient method for solving partial differential equations. *Proceedings of the Second SIMS EUROSIM Conference on Modelling and Simulation, SIMS EUROSIM 2024*. <https://doi.org/10.3384/ecp212.066>
- Ragan-Kelley, J., Barnes, C., Adams, A., Paris, S., Durand, F., & Amarasinghe, S. (2013). Halide: a language and compiler for optimizing parallelism, locality, and recomputation in image processing pipelines. *ACM Sigplan Notices*, 48(6), 519–530. <https://doi.org/10.1145/2491956.2462176>
- Roper Pol, A., Mandal, S., Brandenburg, A., Kahniashvili, T., & Kosowsky, A. (2020). Numerical simulations of gravitational waves from early-universe turbulence. *Physical*

- Review D*, 102(8), 083512. <https://doi.org/10.1103/physrevd.102.083512>
- Steuer, M., Rummel, T., & Dubach, C. (2017). LIFT: A functional data-parallel IR for high-performance GPU code generation. *Proceedings of the 2017 IEEE/ACM International Symposium on Code Generation and Optimization*, 74–85. <https://doi.org/10.1109/CGO.2017.7863730>
- Sujeeth, A. K., Brown, K. J., Lee, H., Rompf, T., Chafi, H., Odersky, M., & Olukotun, K. (2014). Delite: A Compiler Architecture for Performance-Oriented Embedded Domain-Specific Languages. *ACM Transactions on Embedded Computing Systems*, 13(4s), 1–25. <https://doi.org/10.1145/2584665>
- The Astaroth developers. (2026). *Astaroth documentation*. <https://toxypuro.github.io/astaroth>
- Trott, C. R., Lebrun-Grandié, D., Arndt, D., Ciesko, J., Dang, V., Ellingwood, N., Gayatri, R., Harvey, E., Hollman, D. S., Ibanez, D., & others. (2021). Kokkos 3: Programming model extensions for the exascale era. *IEEE Transactions on Parallel and Distributed Systems*, 33(4), 805–817. <https://doi.org/10.1109/tpds.2021.3097283>
- Väisälä, M. S. (2017). *Magnetic Phenomena of the Interstellar Medium in Theory and Observation* [PhD thesis, University of Helsinki]. <https://urn.fi/URN:ISBN:978-951-51-2778-5>
- Väisälä, Miikka S., Pekkilä, J., Käpylä, M. J., Rheinhardt, M., Shang, H., & Krasnopolsky, R. (2021). Interaction of large-and small-scale dynamos in isotropic turbulent flows from GPU-accelerated simulations. *The Astrophysical Journal*, 907(2), 83. <https://doi.org/10.3847/1538-4357/abceca>
- Väisälä, Miikka S., Shang, H., Galli, D., Lizano, S., & Krasnopolsky, R. (2023). Exploring the formation of resistive pseudodisks with the GPU code astaroth. *The Astrophysical Journal*, 959(1), 32. <https://doi.org/10.3847/1538-4357/acfb00>
- Warnecke, J., Korpi-Lagg, M. J., Rheinhardt, M., Viviani, M., & Prabhu, A. (2025). Small-scale and large-scale dynamos in global convection simulations of solar-like stars. *Astronomy & Astrophysics*, 696, A93. <https://doi.org/10.1051/0004-6361/202451085>
- Yokelson, D., Lappi, O., Ramesh, S., Väisälä, M. S., Huck, K., Puro, T., Norris, B., Korpi-Lagg, M., Heljanko, K., & Malony, A. D. (2024). SOMA: Observability, monitoring, and in situ analytics for exascale applications. *Concurrency and Computation: Practice and Experience*, 36(19), e8141. <https://doi.org/10.1002/cpe.8141>
- Zhang, W., Almgren, A., Beckner, V., Bell, J., Blaschke, J., Chan, C., Day, M., Friesen, B., Gott, K., Graves, D., Katz, M., Myers, A., Nguyen, T., Nonaka, A., Rosso, M., Williams, S., & Zingale, M. (2019). AMReX: A framework for block-structured adaptive mesh refinement. *Journal of Open Source Software*, 4(37), 1370. <https://doi.org/10.21105/joss.01370>